

We are aiming for grade C

1 Background

We plan to implement water caustics using photon mapping for an underwater environment. We decided to limit the project to a Cornell box as the environment and use a mesh with a circular wave pattern to mimic the images produced in this work

https://www.researchgate.net/publication/6582416_Caustics_Mapping_An_Image-Space_Technique_for_Real-Time_Caustics

2 Implementation

For this project we will use Unity3D for the rendering. Both of us have some limited prior experience with Unity3D but not specifically for rendering. There seems to be resources¹ for Photon mapping for Unity online. Some of the implementation resources are listed below, this includes two medium articles² with attached source code³ detailing one method, and a youtube video⁴ following another tutorial with relatively unimpressive results.

The linked github code is mostly written in GLSL, which none of us are familiar with and the water surface is not static, which might make evaluation harder.

While looking into which assets to use we decided that for texture painting it would be easiest if we did not include colors in our Cornell box, instead creating a new black texture to paint the light values to.

Unity Water caustics: <https://blog.durablescope.com/post/WaterSimCaustics/>

Using compute shaders in Unity: <https://catlikecoding.com/unity/tutorials/basics/compute-shaders/>

3 Evaluation

We will use the valuation method called comparative evaluation. This technique is often used by software developers by comparing two different programs. For this project however, we will evaluate our project by comparing it to a real life example. Our goal is to re-create an image similar to the rendered and compare the results by different features.

4 Plan

1. Gather information about photon mapping, water caustics, photon gathering and gaussian filter.
2. Create a model for a Cornell box with a single texture and the water in Blender.
3. Import the model and create the environment in Unity.
4. Create an algorithm for the refraction of the light and saving photons in the texture.

¹ <https://github.com/cvryn7/Cornell-Box-Photon-Mapping>

² <https://medium.com/@evanwallace/rendering-realtime-caustics-in-webgl-2a99a29a0b2c>
<https://medium.com/@martinRenou/real-time-rendering-of-water-caustics-59cda1d74aa>

³ <https://github.com/martinRenou/threejs-caustics/>

⁴ <https://www.youtube.com/watch?v=hFY208MusIM>

5. Create the photon gathering and gaussian filter.
6. Test the product and evaluate the results.

5 Contingency plan

Evaluation becomes tricky without comparison and I don't expect that our final version will be in real-time. It might be more useful to instead compare two approaches of caustics rendering, none of which are in real-time. For example, comparing photon mapping methods that are slightly different or comparing photon mapping to ray tracing.